K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

SMART ACADEMIC PERFORMANCE & ANALYTICS SYSTEM (SAPAS)

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UNDER THE ESTEEMED GUIDANCE

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CERTIFICATE

This is to certify that the project based laboratory report entitled “SMART ACADEMIC PERFORMANCE & ANALYTICS SYSTEM (SAPAS) submitted by M.SAI NIKHIL bearing Regd. No. 2500032497 to the Department of COMPUTER SICENCE ENGINEERING , KL University in partial fulfillment of the requirements for the completion of a project based Laboratory in “Computational Thinking for Problem Solving (24SC1102)” course in I B Tech I Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2024 – 2025.

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ABSTRACT

The Smart Academic Performance & Analytics System (SAPAS) is a Python-based application designed to enhance the monitoring and analysis of student academic performance. In modern educational institutions, managing and evaluating large volumes of student data manually is inefficient and often leads to delayed identification of underperforming students. This project aims to address these challenges by automating the collection, processing, and analysis of academic data such as marks, attendance, and assignment scores.

The system is developed using Python and incorporates data analysis and visualization libraries to generate meaningful insights from student records. It provides functionalities such as performance tracking, report generation, and graphical representation of student progress. By using basic predictive techniques, SAPAS can identify trends and forecast student performance, enabling early intervention by educators.

Furthermore, the system offers a user-friendly interface that allows teachers and administrators to easily input data and access detailed performance reports. The integration of analytics helps in making data-driven decisions, improving teaching strategies, and supporting student success.

Overall, SAPAS simplifies academic data management, enhances accuracy, and promotes proactive educational support. It serves as an efficient tool for institutions aiming to improve student outcomes through intelligent performance analysis.

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INTRODUCTION

In today's digital era, educational institutions generate a vast amount of academic data, including student marks, attendance records, assignments, and performance reports. However, many institutions still rely on traditional manual methods or basic systems to manage and analyze this data. These approaches are often time-consuming, error-prone, and lack the ability to provide meaningful insights into student performance. As a result, identifying struggling students and taking timely action becomes difficult.

To overcome these limitations, the Smart Academic Performance & Analytics System (SAPAS) is developed as a Python-based solution that automates the process of tracking, analyzing, and predicting student academic performance. The system utilizes data processing and analytical techniques to transform raw academic data into useful information that supports decision-making for educators and administrators.

SAPAS focuses on improving the efficiency and accuracy of academic monitoring by integrating features such as performance evaluation, data visualization, and predictive analysis. By analyzing patterns in student data, the system can identify trends and highlight students who may require additional support. This enables early intervention and helps improve overall academic outcomes.

The project is designed with a user-friendly interface, making it accessible to teachers and institutional staff with minimal technical knowledge. By leveraging the capabilities of Python and its powerful libraries, SAPAS aims to create a smart, reliable, and scalable system for modern education management.

AIM

The aim of the Smart Academic Performance & Analytics System (SAPAS) is to develop a Python-based intelligent system that automates the collection, analysis, and prediction of student academic performance. The project seeks to transform raw educational data—such as marks, attendance, and assignments—into meaningful insights that help educators monitor student progress effectively.

ADVANTAGES

- Improves accuracy by reducing manual errors in student data handling
- Saves time through automated data collection and analysis
- Helps in early identification of underperforming students
- Provides clear visual reports and insights for better understanding
- Supports data-driven decision making for teachers and administrators

DISADVANTAGES

- Depends on accurate data input; incorrect data can lead to wrong analysis
- Requires basic technical knowledge to operate and maintain the system
- Initial setup and development may take time and effort
- Limited prediction accuracy if advanced algorithms are not used
- Data privacy and security concerns if not properly managed

FUTURE IMPLEMENTATION

- Development of a mobile application for easy access by students, teachers, and parents
- Implementation of real-time data updates and live performance tracking
- Addition of personalized recommendations for students to improve their performance
- Integration with existing school/college management systems (ERP)

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS:

The major software requirements of the project are as follows: Language: Python.

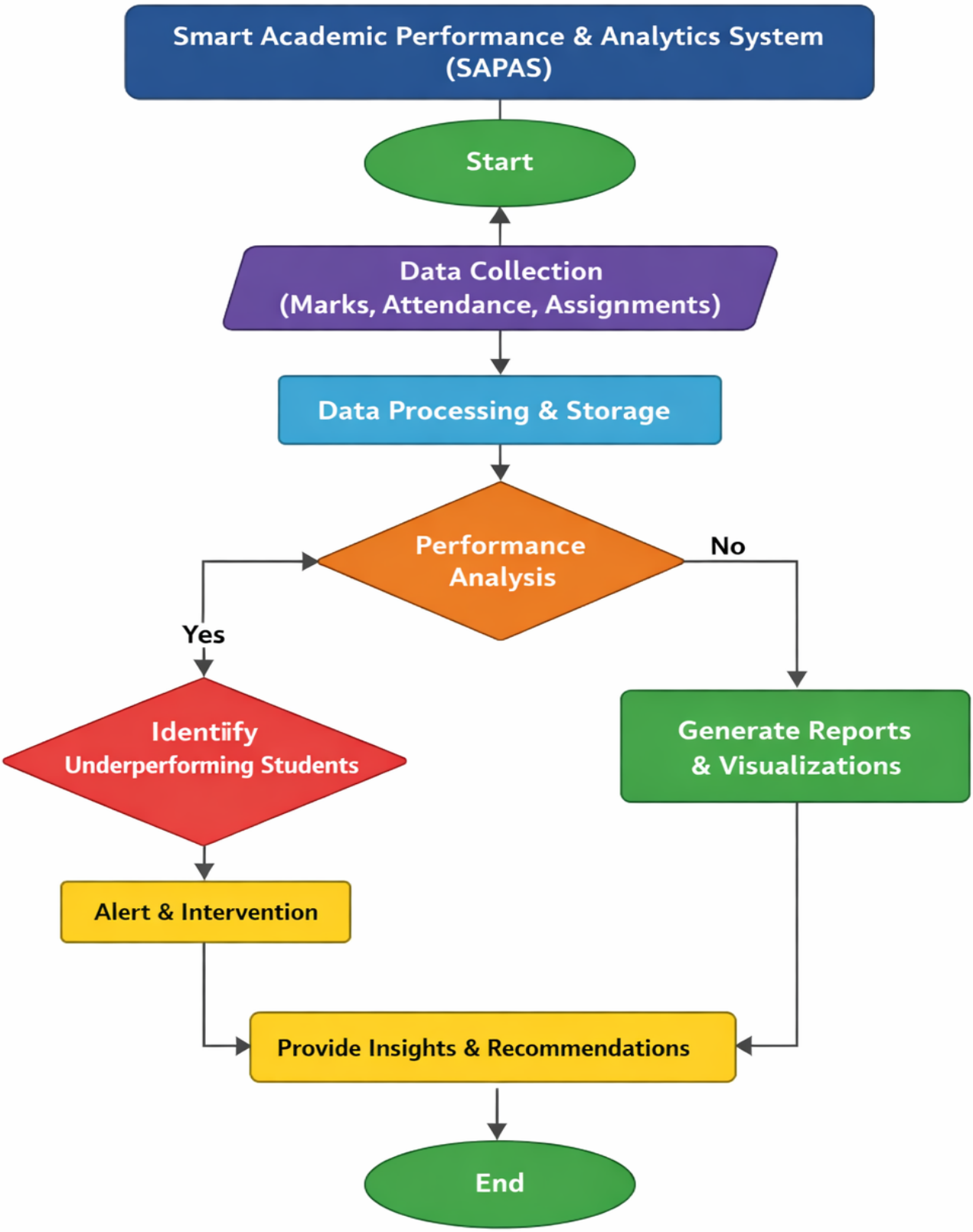
Operating system: Windows 10 or more.

Technical requirements: Monitor, CPU, Keyboard, Mouse, etc...

HARDWARE REQUIREMENTS:

The hardware requirements that map towards the software are as follows: RAM: 8 GB.

Processor: i5/i7


Smart Academic Performance & Analytics System (SAPAS)
Start
Data Collection (Marks, Attendance, Assignments)
Data Processing & Storage
Performance Analysis
No
Yes
Identiify Underperforming Students
Generate Reports & Visualizations
Alert & Intervention
Provide Insights & Recommendations
End

ALGORITHM

Step 1: Start

Step 2: Import required libraries (Pandas, NumPy, etc.)

Step 3: Collect student data

- Input marks, attendance, and assignment scores

Step 4: Store data in a dataset (CSV/Database)

Step 5: Preprocess the data

- Handle missing values
- Convert data into proper format

Step 6: Calculate performance metrics

- Total marks
- Average score
- Attendance percentage

Step 7: Analyze performance

- If average < threshold mark as "Underperforming"
- Else mark as "Good/Normal"

Step 8: (Optional) Apply prediction model

- Train model using historical data
- Predict future performance

Step 9: Generate outputs

- Display student performance
- Create reports and graphs

Step 10: Provide insights

- Suggest improvements for weak students

Step 11:

End

IMPLEMENTATION

```
import json

class Student:
    def __init__(self, student_id, name, marks, attendance, assignments):
        self.student_id = student_id
        self.name = name
        self.marks = marks
        self.attendance = attendance
        self.assignments = assignments
        self.status = "Good"

    def average(self):
        return (self.marks + self.attendance + self.assignments) / 3

    def evaluate(self):
        if self.average() < 50:
            self.status = "Underperforming"

    def to_dict(self):
        return self.__dict__

    def display(self):
        print(f"{self.student_id} | {self.name} | {self.average():.2f} | {self.status}")


def save(students):
    with open("students.json", "w") as f:
        json.dump([s.to_dict() for s in students], f)


def load():
    try:
        with open("students.json", "r") as f:
            return [Student(**d) for d in json.load(f)]
    except:
        return []
def add(students):
    try:
        name = input("Name: ")
        m = float(input("Marks: "))
```

```
        a = float(input("Attendance: "))
        asg = float(input("Assignments: ")
        s = Student(len(students)+1, name, m, a, asg)
        s.evaluate()
        students.append(s)

        print("Added!")
    except:
        print("Error!")


def view(students):
    for s in students:
        s.display()


def search(students):
    name = input("Search name: ").lower()
    for s in students:
        if name in s.name.lower():
            s.display()


students = load()

while True:
    print("\n1.Add 2.View 3.Search 4.Save&Exit")
    ch = input("Choice: ")

    if ch == "1":
        add(students)
    elif ch == "2":
        view(students)
    elif ch == "3":
        search(students)
    elif ch == "4":
        save(students)
        print("Saved!")
        break
    else:
        print("Invalid!")
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
Run    NEW python

C:\Users\Nikhil\AppData\Local\Programs\Python\Python314\python.exe "C:\Users\Nikhil\PycharmProjects\PythonProject\NEW python.py"

1.Add 2.View 3.Search 4.Save&Exit
Choice: 1
Name: nikhil
Marks: 75
Attendance: 75
Assignments: 2
Added!

1.Add 2.View 3.Search 4.Save&Exit
Choice: 2
1 | nikhil | 50.67 | Good

1.Add 2.View 3.Search 4.Save&Exit
Choice: 3
Search name: nikhil
1 | nikhil | 50.67 | Good

1.Add 2.View 3.Search 4.Save&Exit
Choice:
```

CONCLUSION

The Smart Academic Performance & Analytics System (SAPAS) demonstrates how Python can be effectively used to automate the management and analysis of student academic data. By organizing information such as marks, attendance, and assignments, the system reduces manual effort and improves accuracy in evaluating student performance.

The system also helps in identifying underperforming students at an early stage, enabling timely intervention and support. Overall, SAPAS promotes a data-driven approach to education, enhancing decision-making, efficiency, and academic outcomes in modern institutions.